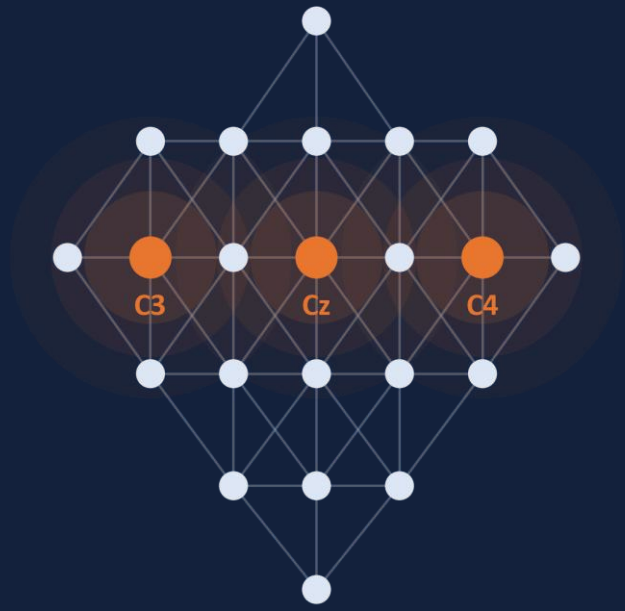
FINAL REVIEW · CSE4006 DEEP LEARNING

Neural Decoding of Motor Imagery from EEG Signals using Deep Learning



Cross-subject decoding on BCI Competition IV-2a, tested on people the model has never seen

Group 5

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github.com/sathvikparangi907-web/neural-decoder-eeg-motor-imagery

RECAP OF THE PHASES

Three phases: problem, design, build

PHASE 1

The problem

Every new user needs a calibration session first. We surveyed the published decoders.

PHASE 2

The design

HCT-Net designed at 20,996 parameters, with a leave-one-subject-out test plan.

PHASE 3

The build

Seven activities built and run. Six decoders compared; the final model has 3,924 parameters.

Objective throughout: decode four imagined movements from the EEG of a person the model has never seen, with no calibration session.

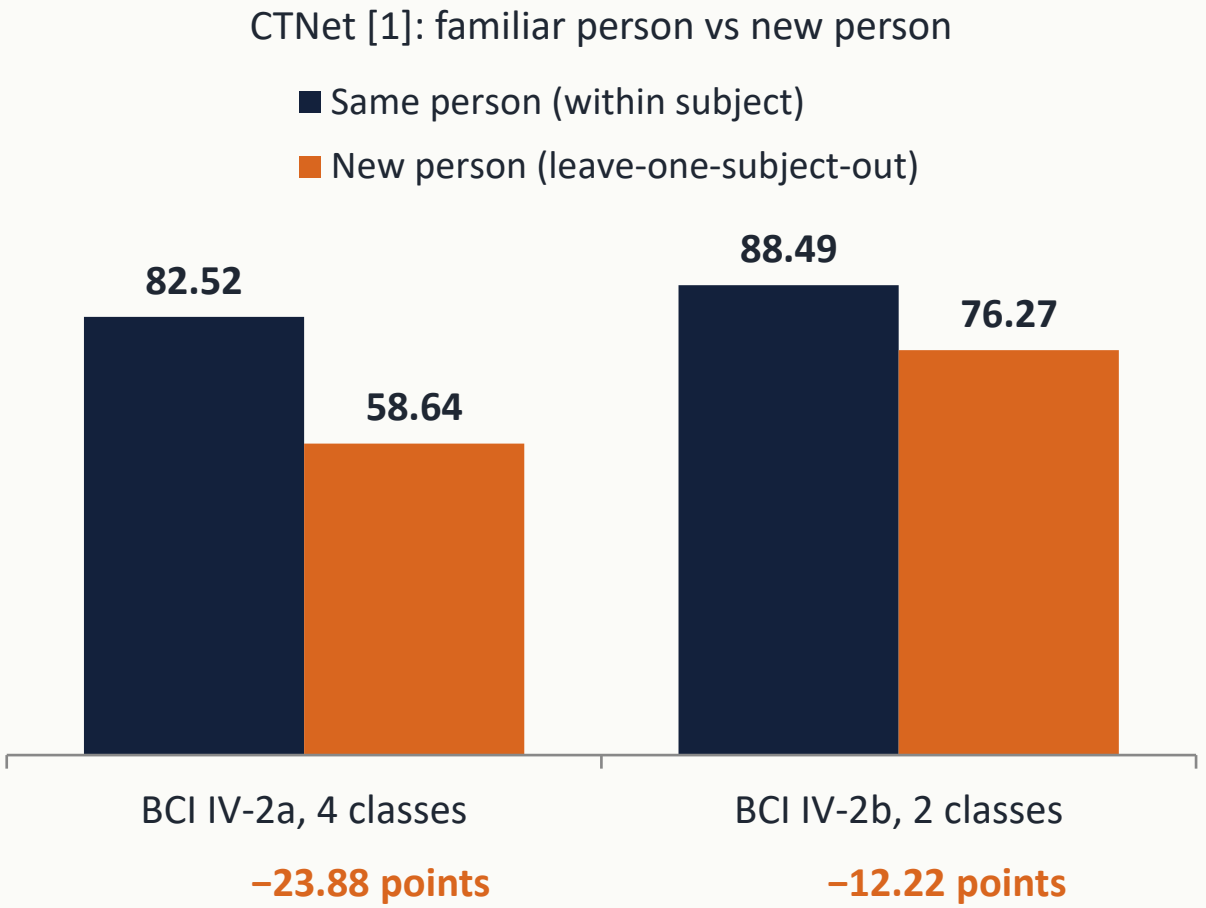
PROBLEM STATEMENT

Decoders fail on people they have not seen

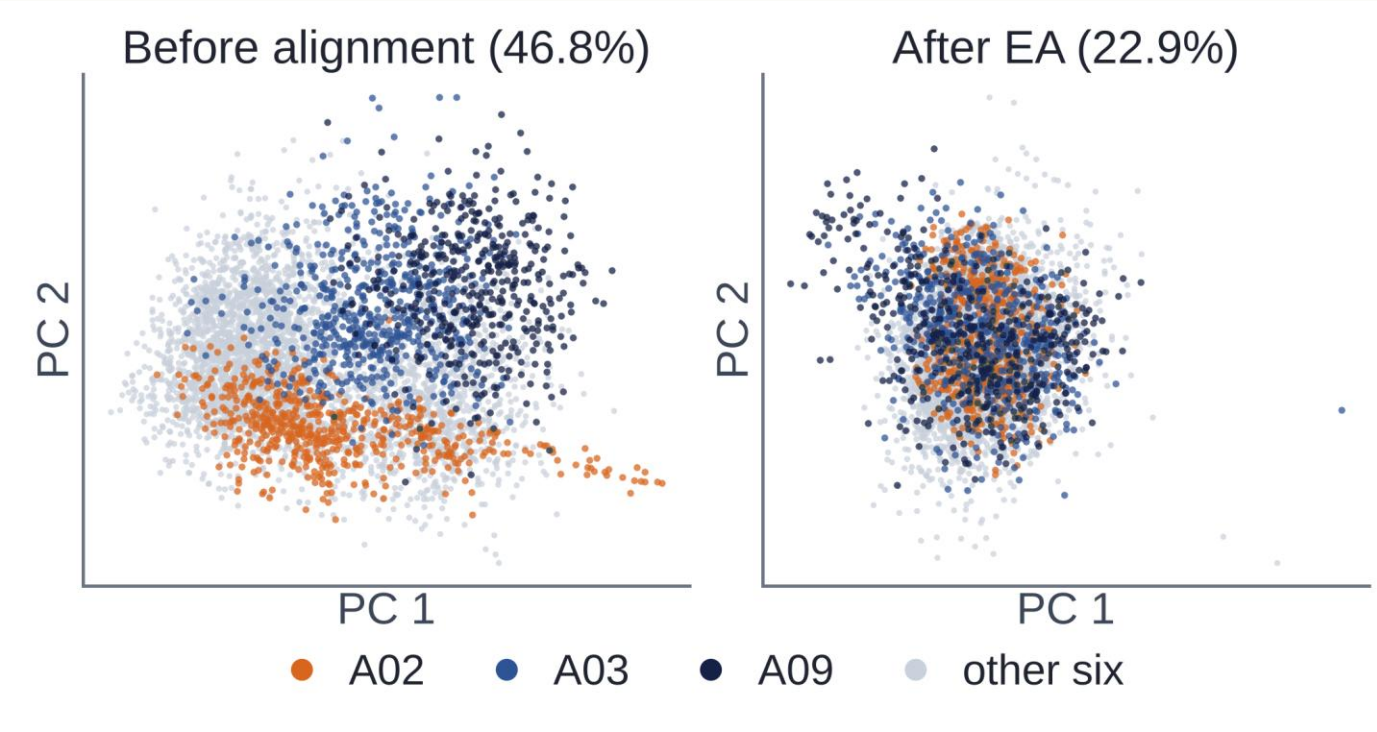
A brain–computer interface turns imagined movement into commands, for people with ALS, spinal-cord injury or stroke.

Today every new user must first record a labelled calibration session.

Goal: a decoder that works for a new person with zero calibration.



Every person's EEG sits in its own place



Each dot is one trial. Left: before alignment. Right: after Euclidean alignment [4].

46.8%

of trials lie closer to their own person's centre than to anyone else's. Chance is 11.1%.

22.9%

after Euclidean alignment of each person's data.

Skull, cortex, cap position and imagery strategy differ, so the signal carries the person as well as the movement.

OBJECTIVES

Six objectives, set before the build

1 Pipeline

A reproducible preprocessing and evaluation pipeline for BCI Competition IV-2a

2 Baselines

EEGNet, ATCNet, CTNet and EEG Conformer, with FBCSP as a classical control, under one protocol

3 HCT-Net

Combine EEGNet's convolutional front end with windowed Transformer attention

4 Unseen people

Leave-one-subject-out testing: every number comes from a person the model never saw

5 Metrics

Accuracy, Cohen's κ, precision, recall and F1, per person and on average, with model size

6 Honest answer

State whether the hybrid helps new people while staying small, whichever way it goes

LITERATURE SURVEY

What the published models do on a new person

Model	Parameters	Main idea	What HCT-Net takes	New person*
FBCSP + LDA (2008) [11]	—	Hand-designed spatial filters per frequency band, then a linear classifier	Classical control only	**not reported**
EEGNet (2018) [7]	3,188	Compact convolutions: frequency, then spatial, then a summary	The whole convolutional front end	**56.85%**
ATCNet (2023) [8]	113,732	Attention and temporal convolution inside sliding windows	Attention limited to short windows	**not reported**
EEG Conformer (2023) [6]	697,412	Convolution for local features, a Transformer across the trial	Convolution first, attention after	**53.41%**
CTNet (2024) [1]	152,364	EEGNet-style front end feeding a six-layer Transformer	The encoder, cut to two layers	**58.64%**

* Published leave-one-subject-out accuracy on BCI IV-2a, from the CTNet paper's re-runs under one protocol [1]. Parameters for our 875-sample input.

Gap: size does not predict accuracy on a new person, and two of the five have no published new-person figure.

OUR NOVELTY

What is new in this project?

WE TESTED	WHAT WE FOUND
Does a larger network help a new, unseen person?	**No reliable benefit from added depth, width or attention**
Does reducing the signal difference between people help?	**Yes: Euclidean alignment and batch-norm adaptation improved transfer**
Can every decision be made without the test labels?	**Yes: the validation person and the test person were kept separate**

The important factor was reducing person-to-person signal mismatch, not making the network bigger.

DATASET

Our data: 9 people, 4 imagined movements

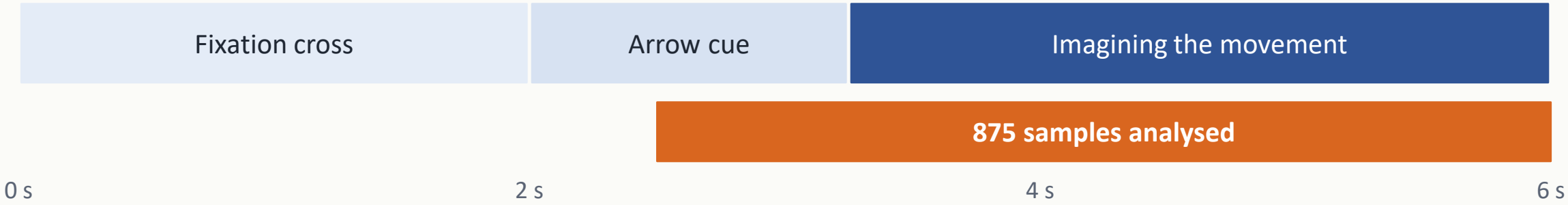
9
people, A01 to A09

22
EEG electrodes

4
movements: left hand, right hand, feet, tongue

5,184
labelled trials, two days per person

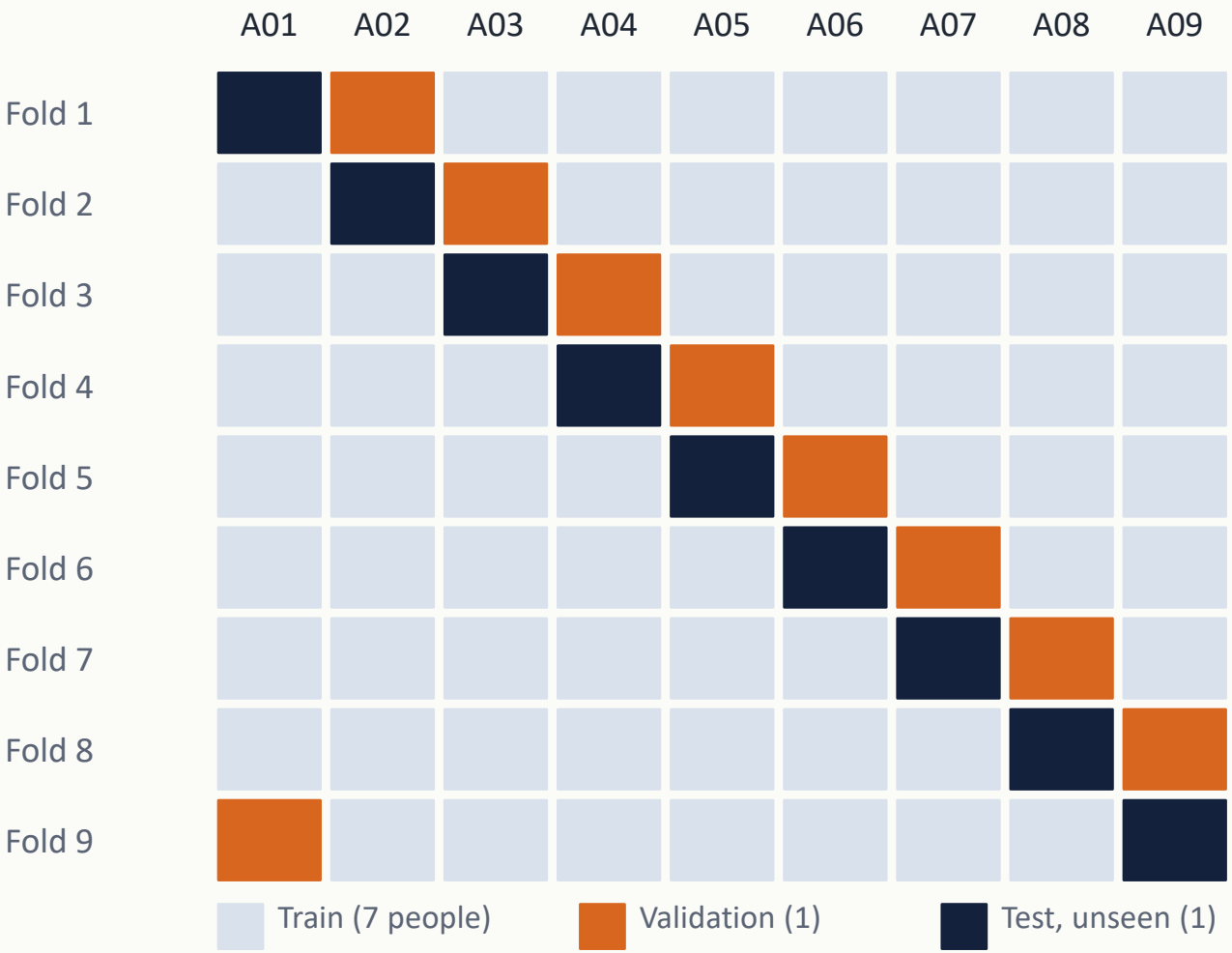
One trial, and the window the decoder sees



Window: 0.5–4.0 s after the cue, leaving out the response to the arrow itself. Chance is 25%.

Brunner et al. [3]; 250 Hz, two sessions per person on different days.

Each person is tested once, as a stranger



7 people train the model.

1 person validates: it chooses between designs.

1 person is tested once, at the very end.

Early stopping uses a 10% stop-set from the training people, and every setting runs with three seeds.

LABEL-FREE PIPELINE

Nothing about the new person is ever labelled

Once

per person, cached

- Keep the 22 EEG channels
- Band-pass 4–38 Hz
- Cut 0.5–4.0 s after the cue
- Flag artefacts above 100 µV

Every fold

training people only

- Euclidean alignment per session [4]
- Standardise with training statistics
- Augment by recombining same-class segments

New person

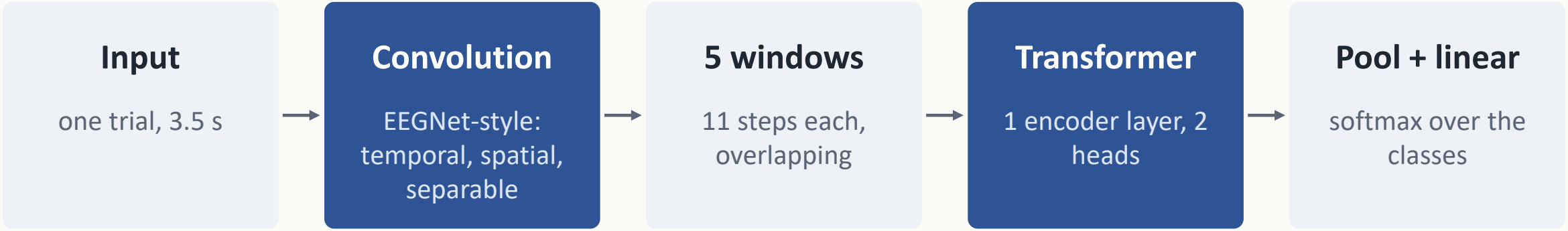
no labels, no retraining

- Align their own unlabelled trials
- Re-estimate batch-norm statistics [5]
- Predict the imagined movement

Zero calibration: both steps that adapt to a new person use only that person's unlabelled trials.

MODEL ARCHITECTURE

HCT-Net A: a compact hybrid model



22 × 875
input: electrodes × samples

3,924
trainable parameters

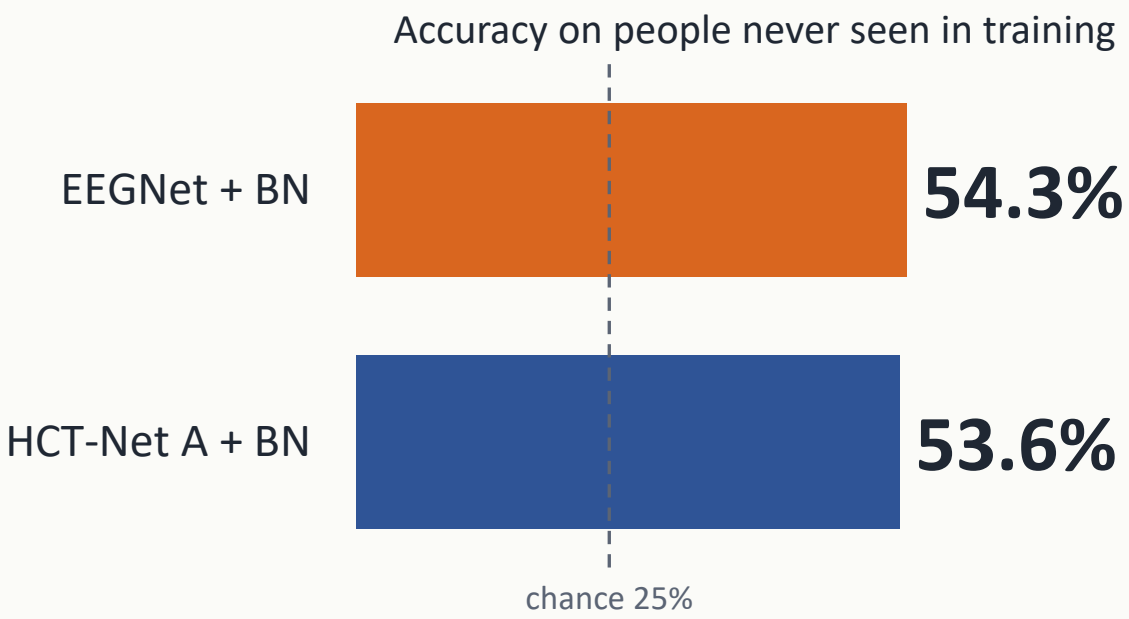
1
Transformer encoder layer

4
classes out

Borrowed building blocks: EEGNet-style convolution [7] feeding a Transformer encoder [9]. The Phase 2 design had 20,996 parameters; a size sweep showed the smallest version did as well.

FINAL RESULT

53.6% vs 54.3%: a tie, not a win



Difference: **0.7 points**
Predefined decision threshold: **0.75 points**
Conclusion: **a tie, not a claimed win**

3,924
parameters in HCT-Net A

κ = 0.381
Cohen's kappa (0 = chance)

9
people, each tested once as a stranger

WHAT HELPED, WHAT FAILED

Alignment helped; extra size did not

Helped

Euclidean alignment [4]

+9.4 points on EEGNet, 9 of 9 people better

Batch-norm re-estimation [5]

+4.3 points on HCT-Net A, 9 of 9 people better

Did not help

5× more parameters (design D)

52.7% vs 53.6%: no gain

Attention

removing it cost nothing (48.8% vs 46.9%, v1)

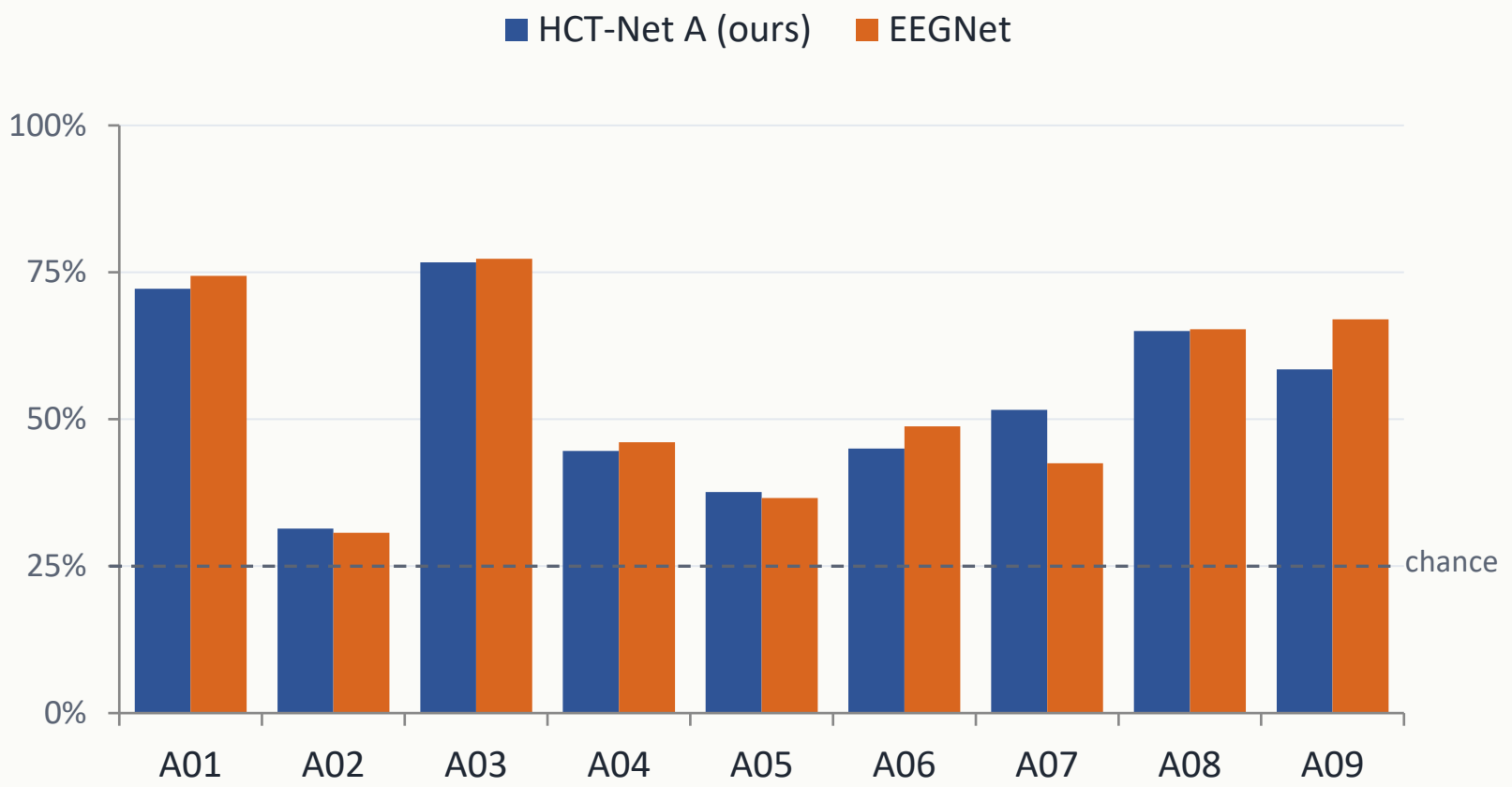
Batch norm inside the encoder

49.3% vs 53.6% on validation, 9 of 9 worse

Protocol v2, three seeds, unless marked v1 (earlier protocol, one seed). Article Tables X, XI, XIV and XVI–XVIII.

PER-PERSON RESULTS

Per person, accuracy runs from 31% to 77%



31–77%

A02 lowest, A03 highest

4 of 9

people below 46%: A02, A05, A04, A06

Feet

is the hardest class (appendix A3)

Test accuracy per person, protocol v2 with BN adaptation, mean of three seeds (article Table XXV).

CHALLENGES FACED

Three challenges, each solved and measured

CHALLENGE 1

The validation person flattered itself

Problem

It stopped training and chose designs.

Fix

Early stopping moved to a 10% stop-set.

Result

Validation–test gap: 6.9 → 0.3 points

CHALLENGE 2

One run was not enough

Problem

Seed noise moved results by about a point.

Fix

Three seeds and a 0.75-point rule.

Result

Two single-run gains failed and were dropped

CHALLENGE 3

The Phase 2 model transferred worst

Problem

46.9%: lowest of the five networks (v1).

Fix

A size sweep and label-free adaptation.

Result

3,924 parameters, 53.6%: level with EEGNet

LIMITATIONS

What this study cannot claim

5.4 points

Nine people is a small sample: differences below about 5.4 points cannot be detected reliably (80% power).

V1 only

ATCNet, CTNet and EEG Conformer have one-seed results under the earlier protocol; they were not re-run under v2.

53.6%

is not ready for daily use: four people stay below 46%, and a familiar person scores 18.5 points higher.

Not size, but alignment and adaptation

Conclusions

- HCT-Net A, 3,924 parameters: 53.6% on unseen people, a tie with EEGNet
- Alignment (+9.4) and label-free adaptation (+4.3) moved results most
- Above 3,924 parameters, extra size bought nothing

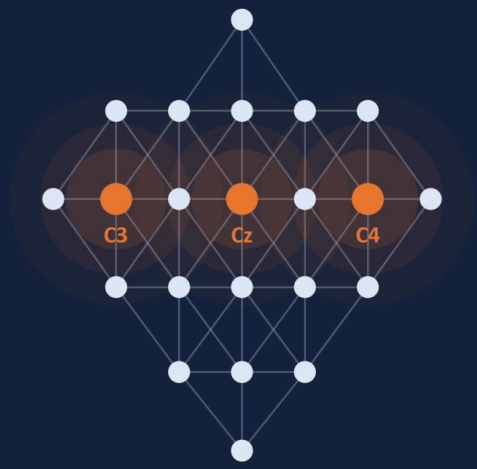
Future scope

- Adapt online as a new user's trials arrive
- Pre-train on PhysioNet MI (109 people), then adapt
- Re-run every baseline under the final protocol

Contribution: the building blocks are borrowed; what is new is the measurement: six decoders in one fair pipeline, a strict selection protocol, and evidence that the cross-subject gap is not a capacity problem.

Bigger models were not the answer.

Better subject alignment was.



Thank you. Questions are welcome.

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APPENDIX A1

All seven Phase 3 activities are complete

01	02	03	04	05	06	07
Setup & data	Preprocessing & EDA	Five baselines	HCT-Net	Training & tuning	Experiments	Evaluation

6.9 → 0.3

points between validation and test once early stopping got its own stop-set

+4.3

test points from label-free adaptation (49.3% → 53.6%); all nine people improve

20,996 → 3,924

parameters after the size sweep, with accuracy level or better

6 decoders

compared in one pipeline, every published figure traced to its source

APPENDIX A2

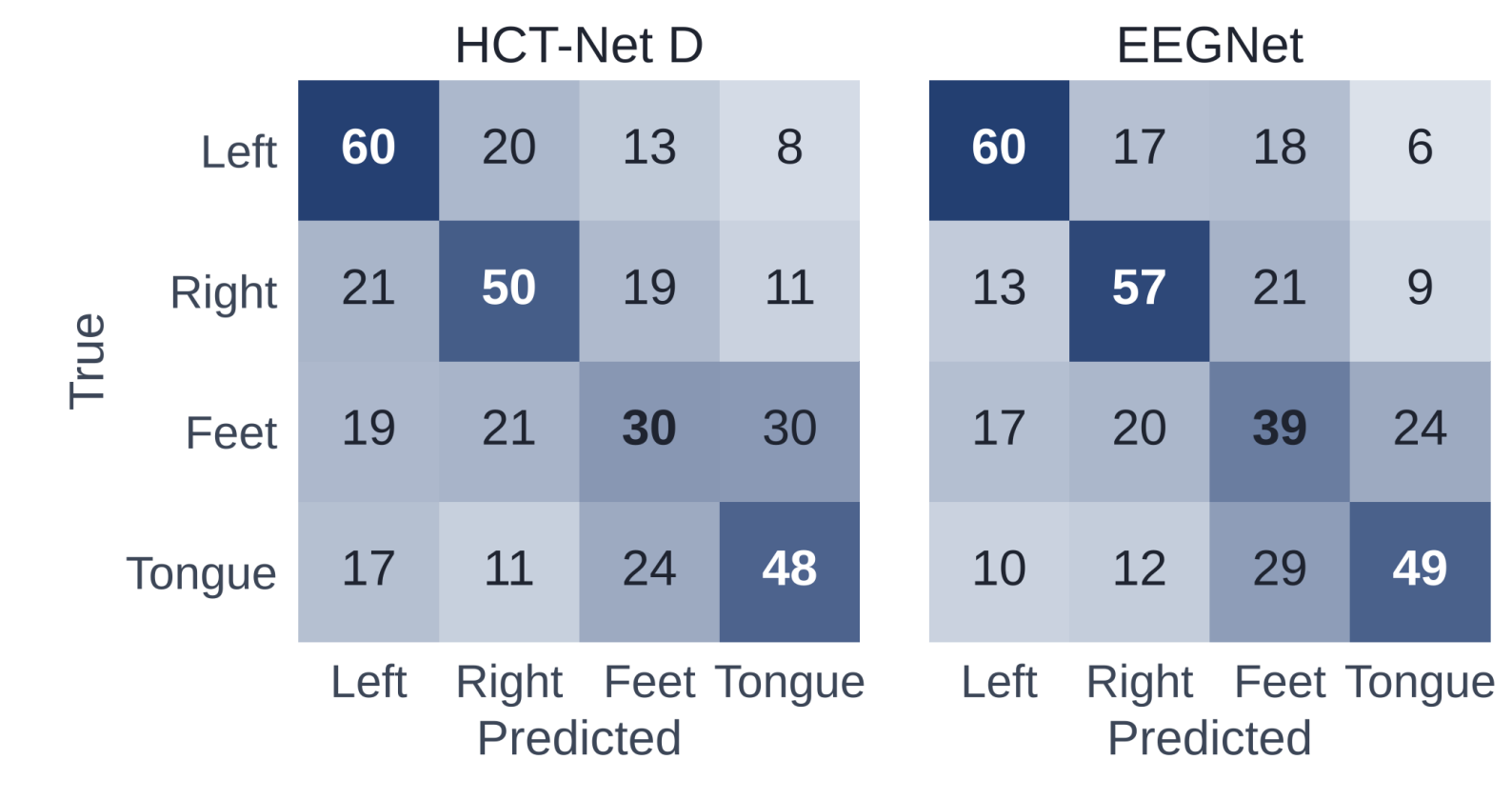
Every model, with its protocol

Model	Parameters	Protocol	Test accuracy	κ
Final protocol v2: three seeds, BN adaptation				
HCT-Net A (ours)	**3,924**	**v2**	**53.6 ± 15.6**	**0.381**
EEGNet [7]	3,188	v2	**54.3 ± 17.0**	**0.391**
HCT-Net D (Phase 2 design)	20,996	v2	**52.7 ± 15.3**	**0.369**
TS-LR + re-centring, classical	—	deterministic	**50.5 ± 15.4**	**0.340**
Earlier protocol v1: one seed, no adaptation (not re-run under v2)				
ATCNet [8]	113,732	v1	**52.4 ± 17.5**	**0.365**
EEG Conformer [6]	697,412	v1	**52.1 ± 14.6**	**0.362**
EEGNet [7]	3,188	v1	**51.2 ± 17.0**	**0.349**
CTNet [1]	152,364	v1	**50.2 ± 13.6**	**0.336**
HCT-Net D	20,996	v1	**46.9 ± 17.5**	**0.292**
FBCSP + LDA [11]	—	v1	**44.2 ± 13.0**	**0.256**

Test accuracy on unseen people, mean ± SD across the nine test people. v1 and v2 numbers are not a single ranking. Article Tables VIII and XVII.

APPENDIX A3

Feet is the hardest class to recognise



Feet is recognised only 30% (HCT-Net D) and 39% (EEGNet) of the time.

Its errors go mostly to tongue: both sit near the middle of the head.

Left and right hand are recognised best.

Percent of each true class, pooled over the nine test folds, protocol v1, one seed (article Fig. 19).

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[8] H. Altaheri et al., “Physics-informed attention temporal convolutional network for EEG-based motor imagery classification,” IEEE TII, 2023.

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[11] K. K. Ang et al., “Filter bank common spatial pattern (FBCSP) in brain–computer interface,” Proc. IJCNN, 2008.